

# LESSON 11.6

## INTERPRETING LINEAR EQUATIONS FIT TO BIVARIATE DATA



### LESSON INTRODUCTION

**MA.8.AR.3.5** Given a real-world context, determine and interpret the slope and y-intercept of a two-variable linear equation from a written description, a table, a graph, or an equation in slope-intercept form.

**MA.8.DP.1.2** Given a scatter plot within a real-world context, describe patterns of association.

### LEARNING TARGETS

- I can describe patterns of associations in data on a scatter plot.
- I can determine the slope and y-intercept from a given line of fit to a set of bivariate data.
- I can interpret the slope and y-intercept of a linear equation in a real-world context.

### BENCHMARK COHERENCE

#### BUILDING ON

**MA.7.DP.1.2** Given two numerical or graphical representations of data, use the measure(s) of center and measure(s) of variability to make comparisons, interpret results, and draw conclusions about the two populations.

#### WORKING TOWARD

**MA.912.AR.3.5** Given the x-intercepts and another point on the graph of a quadratic function, write the equation for the function.

**MA.912.DP.1.2** Interpret data distributions represented in various ways, state whether the data is numerical or categorical, whether it is univariate or bivariate, and interpret the different components and quantities in the display.

### ESSENTIAL QUESTIONS

- What is the purpose of a line of fit?
- How can a line of fit help me understand data?
- How can a line of fit help me to make predictions?

### LESSON NARRATIVE

Students take their exploration of lines of fit a step further by interpreting key components of lines of fit. Given different situations and lines of fit, students work to understand what the slope and y-intercept of a line of fit can tell them about the data. Students also look at if their interpretation is plausible for different situations. Students are guided through how lines of fit aren't exact, but can help us make predictions or generalities about data given in a scatter plot.

## COMPONENT SUMMARY

### 11.6.1 WARM-UP (~5 MIN.)

Critical thinking to complete number puzzles

### 11.6.3 BRING IT TOGETHER (5 MIN.)

Synthesis of how lines of fit can be used

### 11.6.5 YOUR TURN! (5 MIN.)

Independent practice to interpret a line of fit

### 11.6.2 EXPLORATION (10-15 MIN.)

Students explore how a line of fit can assist in understanding the data

### 11.6.4 GUIDED PRACTICE (10-15 MIN.)

Guided through interpreting different lines of fit

### 11.6.6 WRAP-UP (~5 MIN.)

Formative assessment to identify which students may need further assistance

## RESOURCES

### GLOSSARY

Key Terms: N/A

Additional Terms: N/A

### MATERIALS

- (Optional) Highlighters
- (Optional) Graphing utilities for lines of fit

### ADDITIONAL PREPARATION

- Consider using visuals of key terms from this whole unit that show the different relationships, associations, etc.

## TEACHER TIPS

### COMMON MISCONCEPTIONS

- Students may have difficulty interpreting y-intercepts in some of the situations.
- Students may struggle articulating their interpretations of both slope and y-intercept.

### THINGS TO CONSIDER

- This lesson extends the idea of lines of fit to include real-world interpretations.
- It is important for students to understand that lines of fit are approximations and that their interpretations are not always reasonable.

### LITERACY AND LANGUAGE ROUTINES

#### Compare and Connect

In 11.6.4 Guided Practice, consider using the technique of comparing and connecting to further student understanding of the graphs of questions 1 and 2. The line of fit for question 1 extends all the way to the y-intercept, but the line of fit for question 2 does not. Students could discuss the similarities and differences and if it affects their interpretations, specifically for the y-intercepts of the lines of fit.

### MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

#### MA.K12.MTR.6.1 Reasonableness

In this lesson, students further explore lines of fit and how they can be used in conjunction with scatter plots to better understand the data. While 11.6.2 Exploration is scaffolded to help students understand if the interpretations of their lines of fit are reasonable, consider using 11.6.3 Bring It Together to further student understanding of times when lines of fit may lead to reasonable interpretations versus unreasonable by focusing on slope and the y-intercepts.

## DIFFERENTIATE INSTRUCTION

Consider providing Desmos Activity “Put the Point on the Line” as support for remediating understanding of finding slope. Throughout the activity, use student ideas to define slope as a ratio of change in y-coordinates to change in x-coordinates. By the time students get to the end of the activity, they should have a number of ways of talking about slope which will ensure students have a strong understanding as they go into the rest of this lesson.

### 11.6.1 WARM-UP: NUMBER PUZZLES

#### TEACHER MOVES

Consider using question 2 as a discussion starter, asking students to articulate their methods and compare them with other students’ methods.



#### 11.6.1 Warm-Up: Number Puzzles

1. Read the clues for each puzzle to find the answer.

|          | Clues                                                                                                                                                                 | Answer |
|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|
| Puzzle 1 | <ul style="list-style-type: none"> <li>• My value is even.</li> <li>• My value is greater than 6.</li> <li>• My value is less than 10.</li> </ul>                     | 8      |
| Puzzle 2 | <ul style="list-style-type: none"> <li>• My value is even.</li> <li>• My value is not less than 5.</li> <li>• My value is less than 8.</li> </ul>                     | 6      |
| Puzzle 3 | <ul style="list-style-type: none"> <li>• I am a multiple of 10.</li> <li>• I am greater than <math>4^2</math>.</li> <li>• I am less than <math>5^2</math>.</li> </ul> | 20     |

2. What strategies did you use to determine the answer for each puzzle?

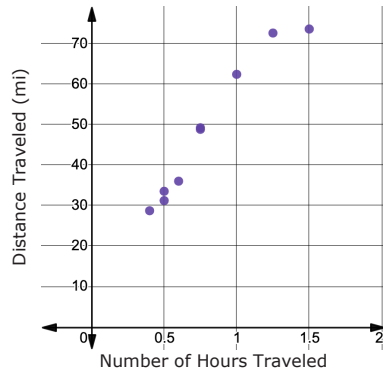
*Answers will vary.*

*Sample response: I used process of elimination to determine the answer.*



### 11.6.2 Exploration: Linear Equations Modeling Bivariate Data

- Camilo traveled with his family from Jasper, FL to Hialeah, FL. The trip was 436 miles using I-95 to avoid tolls. Throughout the trip, Camilo recorded the distance traveled, in miles (mi), for specific amounts of time, in hours. The scatter plot shows the data Camilo recorded.



- The point (0.5, 33.5) is on the scatter plot. Complete the statement to interpret the meaning of this point in context.

For one of the times Camilo recorded their travel, the

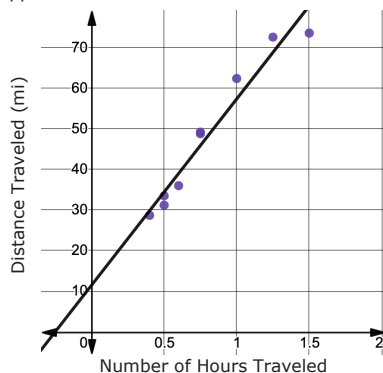
family had traveled  in   
 in

- At what speed was the family traveling during the period of time represented by the point in Part A? **67 miles per hour**

- Describe the association between the number of hours traveled and distance traveled by the family.

**Answers will vary. There is a strong positive linear association between the number of hours traveled and the distance.**

- Suppose the line shown is fit to the data.



The equation of the line of fit is  $y = 45.9x + 11.5$ .

## 11.6.2 EXPLORATION: LINEAR EQUATIONS MODELING BIVARIATE DATA

### ENRICHMENT

Students interchange the labels for the x- and y-axis when completing question 1a. Consider the following questions for scaffolding:

- In the ordered pair, (0.5, 33.5), which is the x-value and which is the y-value?
- Looking at the graph, what is the label of the x-axis?
- Looking at the graph, what is the label of the y-axis?
- How do these labels apply to the coordinate?

### DIFFERENTIATE INSTRUCTION

To assist students with analyzing the graph, consider:

- An enlarged graph or a digital graph
- Auditory descriptions of the line of fit

## 11.6.2 EXPLORATION: LINEAR EQUATIONS MODELING BIVARIATE DATA (CONTINUED)

### TEACHER MOVES

For question 2a, consider listening to student discussions of their understanding of the meaning of line of fit. Encourage students to refer to the definition of line of fit included in prior lessons. A line of fit estimates the relationship between two sets of data. This line represents the approximate relationship between number of hours traveled and distance traveled by the family on the road trip.

### ENRICHMENT

Question 2c may be a challenge for students. Consider using this as an opportunity to listen to their discussions. They aren't expected to write anything here or be correct right away. It's important to give them time to process this idea. They will go deeper into these concepts in Algebra 1. The purpose of this lesson is to have students relate what they already know about linear relationships from Units 7 and 10 in the context of scatter plots.

The y-intercept doesn't make sense in this context because they would not have traveled 11.5 miles in 0 hours. They would travel 0 miles in 0 hours.

#### Optional clarification to discuss in debrief:

The line of fit estimates the relationship so it is approximating the rate of change between the variables and the initial value. The real-world data values may be different than the line, but the line still models the overall trend.

- a. Discuss with your partner what you think this line of fit represents in this context. Summarize your discussion.

*Answers will vary.*

*Sample response: This line represents the approximate relationship between the number of hours traveled and distance traveled by the family on the road trip.*

- b. Complete the statement.

The slope of the line fit to the data is ☐ 11.5. ☒ 45.9. This

means, on average, the family traveled about

☐ 11.5 ☐ miles per hour  
☒ 45.9 ☐ hours per mile on their trip from

Jasper to Hialeah.

- c. Discuss with your partner whether you think the y-intercept of the line makes sense for this context.

*Answers will vary. I do not think the y-intercept makes sense because the family would not have traveled 11.5 miles in 0 hours.*

**11.6.3 Bring it Together: Linear Equations  
Modeling Bivariate Data**

When a linear equation is fit to bivariate data in a scatter plot, the line approximates the relationship between two sets of data to represent the overall trend.

The slope and y-intercept of the line can be interpreted in context, but there are times when the interpretation may not make sense in context because it is representing an estimate based on a set of data.

**11.6.3 BRING IT TOGETHER: LINEAR  
EQUATIONS MODELING BIVARIATE DATA****TEACHER MOVES**

After completing the Bring It Together, consider opening a discussion about any contexts that might not make sense, focusing on the y-intercept. For example, in the question above, it's impossible for the family to have traveled that many miles at a time of 0. What might be some other situations where that would be similar?

## 11.6.4 GUIDED PRACTICE: INTERPRETING LINEAR EQUATIONS FIT TO BIVARIATE DATA

### TEACHER MOVES

#### Compare and Connect

Consider using the graphs from questions 1 and 2 to compare lines of fit. Discussing physical features can lead into discussing the interpretations of both, helping to understand when an interpretation may not make sense.

### DIFFERENTIATE INSTRUCTION

Consider displaying or having a guide for students with the description of the terms in an equation in slope-intercept form.

### ENRICHMENT

To help students distinguish why the interpretation of the y-intercept is unreasonable:

- If Aditya had 0 clients, how much money would you expect him to have in tips? Why?

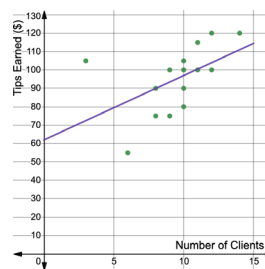
To assist students in completing the interpretation in (a):

- Describe the association of the scatter plot. Is it positive or negative?
- If it's negative, what happens to the points as the y values increase?



### 11.6.4 Guided Practice: Interpreting Linear Equations Fit to Bivariate Data

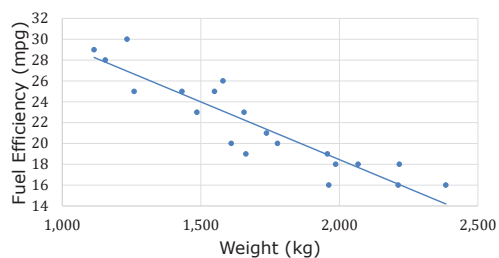
1. Aditya works as a hair stylist and was looking at his earnings for the past few weeks. He created the scatter plot shown to see if there was a relationship between how many clients he saw in a day and how much he earned in tips.



The equation of the line of fit is  $y = 3.5x + 61.94$ .

- a. What is the slope of the line fit to the data?  
**3.5**
- b. Interpret the meaning of the slope in context.  
*Answers will vary. On average the amount of money that Aditya earns in tips increases by \$3.50 for each additional client.*
- c. Explain why the y-intercept of the line estimating the relationship between the variables does not make sense in this context.  
*Answers will vary. The y-intercept of the line does not make sense in this context because Aditya would not have earned \$61.94 after having 0 clients.*

2. The scatter plot shown represents the relationship between a car's weight, in kilograms (kg), and its fuel efficiency, in miles per gallon (mpg).



The equation of the line of fit is  $y = -0.012x + 41.3$ .

- a. Complete the statement.

The ☐ weight ☒ fuel efficiency of the car ☐ increases ☒ decreases by about ☐ 0.012 ☐ 41.3 ☐ kg ☒ mpg as the ☐ weight ☒ fuel efficiency of the car ☐ increases ☒ decreases by 1 ☐ kg. ☒ mpg.

- b. Describe how the direction of the association of the data relates to the slope of the line fit to the data.

*Answers will vary. There is a negative association between the weight and the fuel efficiency, so the slope of the line would be negative.*

## 11.6.4 GUIDED PRACTICE: INTERPRETING LINEAR EQUATIONS FIT TO BIVARIATE DATA (CONTINUED)

### DIFFERENTIATE INSTRUCTION

To assist struggling students, consider using the data you gathered yesterday in regard to students who are struggling to put together a small group to work through question 2 specifically.

## 11.6.5 YOUR TURN!

## TEACHER MOVES

Monitor which representations students are using when on their own to determine the key features to answer the comparison questions. Their approach might illuminate potential gaps or learning styles.

## ENRICHMENT

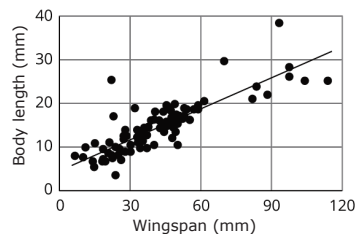
Students may incorrectly identify the y-intercept as the slope. Monitor student work. Guidance can be offered as:

- The slope is the rate of change – which of these values represent a change? How can you tell?
- The y-intercept could be considered as a starting point – which of these values doesn't affect x values?



## 11.6.5 Your Turn!

1. Researchers have collected data comparing the wingspan and body length of butterflies, in millimeters (mm). The scatter plot represents the data collected with the line,  $y = 0.238x + 4.642$  fit to the data.



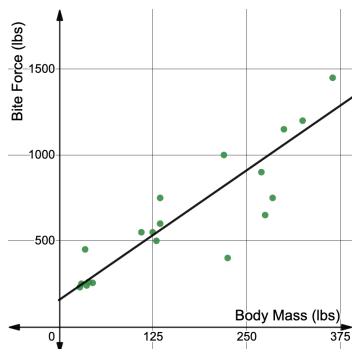
Interpret the slope of the equation modeling the relationship.

*Answers will vary. The slope of the equation tells us that on average the body length of a butterfly increases by 0.238 mm for every 1 mm increase in wingspan.*



### 11.6.6 Wrap-Up: Ticket Out the Door

1. Zoologists studied different species of animals to examine if there was any relationship between the body mass of the animal, in pounds (lbs), and the animal's bite force, in lbs.



The equation of the line of fit is  $y = 3.02x + 154.7$ .

- a. What is the slope of the line fit to the data?
- b. Interpret the meaning of the slope in context.

3.02

Answers will vary.

Sample response: The slope tells me that an animal's bite force increases an average of 3.02 lbs for each additional pound of body mass.

## 11.6.6 WRAP-UP: TICKET OUT THE DOOR

### DIFFERENTIATE INSTRUCTION

Consider providing students who are struggling to articulate their interpretation sentence starters such as, "As body mass increases ..."

### QUESTIONING

For students who finish early, consider asking students to go through the graphs from this lesson and describe all the associations (positive, negative, strong, weak) as a way to connect to slope and review previous concepts.